

Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

- A. Radial artery
- B. Ulnar artery
- C. Princeps pollicis artery
- D. Brachial artery

Reasoning + Answer:

<thought>

To solve this problem , we need to identify the artery in the anatomical snuffbox region that is most likely damaged , given the teenager's situation with bleeding from the hand .

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<answer>

A . Radial artery </answer>

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The anatomical snuffbox is the depression on the back of the hand between the second , third , and fourth fingers and the thumb . It is formed by the tendons of the long and short extensor pollicis brevis and brevis muscles , the long and short abductor pollicis brevis muscles , and the scaphoid and tubercles of the radius .

The main arteries passing through this region are the superficial palmar arch (anastomosis between the superficial branch of the ulnar artery and the terminal branch of the radial artery) .

In a penetrating stab wound causing significant bleeding , the artery most likely involved is the terminal branch of the radial artery , which directly supplies the skin of the back of the hand , particularly in the anatomical snuffbox .

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